

Kadhir Ponnambalam

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Technical Skills: FPGA, Machine Learning, Verilog, Python, Digital Logic
Core Strengths: Technical Documentation, Systems Thinking, Team Collaboration

EDUCATION

Bachelor of Engineering Co-op (BEng) | Computer Engineering | 2025 - 2029 **GPA: 12.0**
McMaster University - Entrance Scholarship \$3500, Edwin Marwin Dalley Memorial Scholarship

WORK EXPERIENCE

Summer Research Intern **Jun 2026 – Present**

Nano & Optoelectronics Lab | McMaster: Dr. Jamal Deen & Dr. Shiva Kumar

- Implemented algorithmic changes to existing Cognitive Dynamic System for greater prediction capability as well as higher clinical separability of arrhythmia diagnosis
- Grounded all algorithmic changes in data analysis and presented novel additions to existing pipeline stages for significantly improved accuracy
- Conducted experiments to capture accuracy, AUC, ROC, F1, Precision, Recall scores of novel methods for comparison with current literature

Research Assistant **Jun – Sept 2023**

Micro & Nano Systems Lab | McMaster: Dr. Jamal Deen

- Conducted literature review on current flexible temperature sensing materials to find gaps for investigation
- Developed a novel graphene-based temperature sensor, measured using microcontrollers and electronic components

PUBLICATIONS

- 1) Taheri, M., **Ponnambalam, K.**, Shboul, A. A., Ketabi, M., & Deen, M. J. (2025). Novel Nickel Oxide/Graphene Composite Sensor: A Low-Temperature Approach to Flexible Temperature Sensing. *Advanced Materials Technologies*, e00060.
- 2) Taheri, M., **Ponnambalam, K.**, Al Shboul, A. M., Ketabi, M., & Deen, M. J. (2025). Nickel oxide/graphene-based temperature sensor: Flexible design with low processing temperatures and high stability. *Sensors and Actuators A: Physical*, 116693.

PROJECTS

FPGA based Smart Traffic Light System **Jun - Jul 2026**

[\[https://github.com/kadgitub7/FPGA-AI-Accelerated-Traffic-Controller\]](https://github.com/kadgitub7/FPGA-AI-Accelerated-Traffic-Controller) [\[https://www.youtube.com/watch?v=FnLT4PYCGY\]](https://www.youtube.com/watch?v=FnLT4PYCGY)

- Designed intelligence traffic light system which utilizes Computer Vision (YOLOv8n) for car detection, FPGA (Basys 3) for light decision logic, and Arduino for embedded control of traffic lights (LEDs)
- Smart system achieved ~77% decrease in average wait times over 11 worst case scenarios compared to baseline model with FPGA components measuring Worst Negative Slack = 3.104, True Negative Slack = 0.000, Worst Hold Slack = 0.082, True Hold Slack = 0.000 and power consumption of 0.076 W

FPGA AI Acceleration for Arrhythmia Diagnosis

Mar - Jun 2026

[\[https://github.com/kadgitub7/FPGA-x-CDS-NI\]](https://github.com/kadgitub7/FPGA-x-CDS-NI) [\[https://www.youtube.com/watch?v=AOe1tF_W6hg\]](https://www.youtube.com/watch?v=AOe1tF_W6hg)

- Configured Cognitive Dynamic System python model into a Finite State Machine run on FPGA architecture
- Computed diagnosis for 4 patients in parallel and achieved 90% reduction in power consumption compared to laptop CPU and 70% reduction in computing time compared to python script per patient